Concept2Cure Data Room enhancement delivery plan

14 October 2026. Products: biotechnology, pharma, biologics, medical devices and diagnostics. Markets: USA, EU/EEA, Japan, Canada and Australia, with a separate UK track. Platform: Concept2Cure and AnA 1.0 RI.

Assessment baseline: `concept2cure-v2`, `87b27c897c5d1be20b0527009bcaf425ebbf3af6`. All implementation below is local work on that branch. Research, proposed contracts, completed tests and operational qualification are distinguished throughout this plan. No deployment or agency transmission occurred.

Product direction

Make the Data Room the scientific evidence foundation of the existing platform: preserve original records, interpret results in their scientific context, retain accountable decisions, and carry those relationships into Authoring and filing preparation. AnA proposes and explains; deterministic services calculate and check; named reviewers decide scientific applicability and release.

The advantage to pursue is reliable evidence reuse and change assessment across a product's lifecycle. A percentage appearing in a PDF is not enough: a claim must identify the study or assay, population or specimen, measurement, method, unit, denominator and relevant configuration. A citation must resolve to the precise version and passage or calculation. A change should identify affected claims and obligations while preserving the historical filing.

Competitors already offer AI drafting, citations, global dossier reuse, structured regulatory data and impact assessment. The five hypotheses below require direct comparison, rather than claims that competitors lack them. Defensibility would accumulate through expert-approved scientific relationships, qualified methods, difficult regression cases, reliable integrations and demonstrated client outcomes.

Seven coordinated contributors

Six specialist agents and the lead worked in separate scopes on the canonical branch. They covered current data management and storage; scientific catalog/AnA behavior; retrieval and durable workflow design; filing implementation; regulatory requirements; and competition/customer benchmarks. The lead integrated their findings, implemented filing-pack byte binding, and coordinated an independent review of the integrity changes.

The main study contains baseline features, code evidence, competitor sources and priorities. Detailed handoffs are preserved as repository documents:

- Scientific fact contract and 20 expert fixture groups.

- Relevant-passage receipts, authorization and restartable AnA work.

- Current filing matrix and F0–F9 delivery milestones.

- Regulatory route packs, lifecycle matrices and qualification.

- Five moat hypotheses and customer benchmark protocol.

Completed local implementation: artifact integrity

Both local and S3 storage now recompute loaded-byte SHA-256 and check recorded length before returning an owned object. Missing or malformed digest/length, changed bytes and changed length produce a typed integrity error. Existing tenant/path isolation and missing-object behavior are preserved.

The IVDR download route independently binds PDF, DOCX, ZIP and manifest bytes to the selected tenant's pack-row digest and length. This catches joint file-and-sidecar changes that pass storage self-consistency.

It validates the requested format and emits the checksum of the actual response bytes. A missing original manifest is refused rather than replaced with another snapshot rendering. For legacy packs without an original manifest, the tenant-scoped snapshot must match both recorded canonical hashes before its JSON view is rendered.

This work moves the integrity portion of launch row D2; it does not close the whole launch row. It adds no schema, dependency, model call or new product surface. Signed-URL downloads are outside this byte-returning read path. Audit emission retains its existing best-effort behavior, so this patch does not establish guaranteed durable download auditing or proof of client receipt.

Evidence: D2 contract, failing cases and passing results. The combined storage/download suite passes **106 tests across six files**. Focused TypeScript checks pass. Changed storage files/new tests have clean ESLint; the route retains its four pre-existing warnings with no new warning. An independent agent found no blocking integrity or async-control-flow issue. S3 transport and route database access were mocked; real cloud/PostgreSQL deployment qualification remains separate.

Completed continuation: validation classification

The continued D7 slice now prevents a clean internal FDA subset from clearing a required-production external-validator gate. Explicit server-selected adapter kind, selected/report engine identity and report execution/count/pass coherence replace the old non-noop predicate. Unknown error counts remain unknown, with a validity marker accompanying the retained numeric field.

Advisory policy remains nonblocking, while failed, invalid and unassessed results retain their actual status through the assessor, readiness and market screens, AnA context, deterministic QC warnings and pre-transmit checks. MCP reports adapter discovery separately from execution or qualification. These changes migrate existing consumers; they do not add another validator or alter signing/Shadow Review controls.

The final combined Node 22 run passes **230 tests across 15 files**. Selected-source TypeScript checks pass, and scoped lint adds no warning after the final MCP helper correction. Independent runtime/compound/pre-transmit checks pass 34/34. D7 contract, source scopes, red/green evidence and limits. This is a classification/status fix, not authenticated provider coverage, exact-package report binding, licence qualification or agency acceptance.

The parallel processing-recovery preparation executes actual processing SQL and selected real migrations in embedded PGlite: five intentionally failing contract assertions and three passing controls. It reproduces lost recovery after partial persistence, missing canonical source-byte binding, stale superseded commits/candidates and completion reporting of failed extraction. Successful derived persistence updates Data Room full-text search atomically; there is no separate asynchronous Data Room index stage to invent. That historical preparation preceded the recovery implementation below; its opt-in probe is excluded from normal tests and is not the current regression suite.

Completed continuation: processing recovery

New captures remain pending until verified derived text commits. The canonical processor binds private byte snapshots to the exact capture and recorded original upload, including digest, storage path, program/workspace and extraction behavior. Currentness, live project, disposition and recorded lineage are rechecked at acceptance. A durable stage in existing metadata records fenced attempts, a 15-minute database-time lease, three automatic attempts and bounded backoff; an explicit authorized retry preserves attempt history. There is no new queue, table, index or production dependency.

Failures preserve the capture/edit/adoption receipt and report a named processing outcome. Capture connections are released before the processor acquires its separate transaction connection. The existing screens and AnA responses rely on actual nonblank persisted text, preserve retained eligible data, distinguish historical superseded sources and never equate a completion label with readable evidence. PostgreSQL maintains the existing full-text expression index with the derived write.

The existing bounded sweep now separates runnable work from blocked/busy/not-due/exhausted diagnostics. The tenant-scoped operator command validates arguments before database imports, defaults

to report-only, and outputs whitelisted codes rather than private errors. No scheduler or actual client repair was run.

Independent PostgreSQL sessions reproduced an adjacent withdrawal snapshot defect and verified its correction. Confirmation retains the existing reservation order, rereads live program authority and current impact, reserves export-history fingerprints, and refuses an obsolete preview. The final combined run passes 532 tests across 38 files; 21 independent checks and 13 genuine PostgreSQL cases pass with matching final source hashes. Local source/claim/withdrawal races and a synthetic non-owner RLS case are distinct from deployed-role or production workload qualification. Recovery contract, final verification and limits.

These changes improve D2's integrity/readability/recovery portion. Cold audit-history archival coherence, a static pre-COMMIT adoption-loader pool concern, an unreproduced filing-writer snapshot concern and broader launch acceptance remain separately tracked. Scientific approvals and regional filing qualification are unaffected.

Completed continuation: scientific source preview

The existing AnA catalog tool now has an explicit read-only verification-preview operation, gated by `ANA_CATALOG_SOURCE_FACT_PREVIEW=1` and off by default. Five typed scientific families bind the source-written value and supported context to exact document/original-record/extraction hashes and quotes. Clinical interpretation reuses the existing engine; the first CMC/device/IVD form is an explicit labelled source row. Missing, ambiguous, remote-header, OCR and unsupported language/layout context remains unresolved. Nothing is stored or accepted by this operation, and scientific qualification remains unassessed.

Existing catalog writes now use private proposal snapshots, the initially admitted programme and exact stored extraction, followed by fresh transactional actor/private-project/source checks and one actual UPDATE receipt before chained audit. AnA windows/full-read gates use real stored text length. Discovery and catalog reads require matching recorded source ownership and live parent scope. The current panel and recall preserve legacy human history while disclosing value-only checks and unrecorded context. The four catalog/read/list/search tools return safe unexpected-error messages with unconfirmed outcome, rather than exposing internal details or inventing rollback.

Independent source fixtures exposed and verified corrections for swapped contextual identities, lost bounds/ranges, clipped anchors, omitted source qualifiers, remote clinical headings and malformed prototype names. Genuine PostgreSQL sessions exercise source moves/withdrawal/supersession/re-extraction, current authority, zero-row updates and audit rollback. The scientific checkpoint’s combined Node 22 run passed 1,752 tests across 87 files; 52 independently authored held-out checks and 15 genuine PostgreSQL cases pass with stable final hashes. The 70 scientific fixture requests are included in the combined run, not additional tests. Scientific catalog contract, final verification and limitations.

This is the source-verification foundation of the earlier five-family specification. Typed fact persistence, immutable per-fact decisions, expert scientific applicability and lifecycle/export/purge remain separate work. At the end of this preview increment, read receipts remained aggregate byte-bound evidence and private-project recall had not yet been qualified. The source-access continuation below adds current-actor checks without promoting legacy history into durable delivery proof. No experimental path was enabled for clients.

Completed continuation: AnA source access

The existing AnA document tools, selected Vault drafting sources, lexical/meaning/passage adapters and source-file recall now use canonical current-actor project access. Open-project conversations remain in their admitted project; organization-level discovery includes only accessible unique project/program anchors. SQL applies that scope before counts, rank and LIMIT. Fully unbound chat captures require both the persisted capture creator and upload owner; every recorded private parent remains enforced. Historical by-ID reading is disclosed without new current-source receipts, while retained eligible text preserves its original-file-unavailable notice.

Both chat bootstrap paths and explicit recall carry authenticated user identity and the admitted project reference. Missing or unavailable source recall produces an incomplete-recall notice. Reads recheck access and exact extraction after coverage awaits; both loop hosts recheck pending source snapshots before staging new source context. The existing receipt writer repeats current actor/source/text checks in a short transaction, validates the actual INSERT and handles cancellation. A later observed source refusal withdraws pending Vault reads in that round, while earlier actual commits retain their legacy-history meaning. Tool-step title enrichment uses the same accessible source scope.

Final verification passes **2,927 tests across 119 files** on unchanged code/test hashes, **50 independent held-out checks** and **12 genuine PostgreSQL cases** with three observed blocking barriers and verified fixture removal. The independent corpus combines 39 reviewer-owned and 11 separately authored lead controls; owner totals overlap the combined regression. Scoped diagnostics have zero target errors. Existing lint warnings and one added streaming follow-up complexity warning remain; the work is not a full-repository build or deployed qualification.

This is a locally verified source-access foundation for reliable client-specific evidence reuse. Persisted receipts remain host-budget-admitted legacy byte history. They do not persist extraction/actor/delivery checkpoints or prove provider service. Prior conversation replay, memory/profile policy, all other platform/MCP source readers, deployed RLS and continuous remote-egress revocation are outside this increment. Viewer text access preserves the canonical read policy; the existing receipt INSERT policy is unchanged and a denied write is not counted. The scientific preview remains off. Source-access contract, final proof and boundaries.

Completed continuation: durable source admission

One new immutable public/integer-tenant ledger records exact host-budget admission metadata: server admission ID, current actor/role, original and extraction hashes, bounded UTF-16 ranges with slice hashes, and the unchanged payload hash. The shared V2 descriptor supports one to eight disjoint ranges; the current offset reader adapts its single window. Identical retries preserve identity and report an existing record; changed identity refuses. The writer repeats canonical source/actor checks and acknowledges only after actual COMMIT. Optional legacy insertion uses a savepoint, with narrowly classified permission denial disclosed without claiming legacy history was written.

Actor-aware history filters actor/programme scope before count and LIMIT and re-admits current access. It survives source removal as metadata only. The complete tenant export includes the table; the canonical controlled purge checks current status, retention, pinned final-export evidence and legal holds after source-table reservations, with deletion/status/audit in one transaction. A zero or unconfirmed final status-update acknowledgement rolls back erasure. Migration, runtime privileges and boot immutability checks are registered on the existing canonical paths. The flag `ANA_DURABLE_SOURCE_READ_RECEIPTS=1` remains off by default.

The final combined continuation passes **3,303 tests across 134 files**, **32 independent held-out checks** and **16 genuine PostgreSQL cases**, with stable final hashes and ten observed blocking checkpoints using an ordinary runtime role. Selected-root type checking has zero errors; lint retains one existing offboarding file-size warning. Synthetic HMAC sealing, source/lifecycle prerequisites and application seams are explicit. The full provisioned-schema purge gate remains unqualified; its self-test and the actual new-table export/purge behavior are recorded separately. The checkpoint is host admission. Provider dispatch/delivery, earlier-context replay, authoring/save/export range receipts, structural retrieval and expert scientific decisions require separate contracts. No provider code or calls are included. Durable receipt contract, proof and limits.

Prioritized delivery backlog

These are ordered, bounded increments, not a commitment to ship every proposed capability simultaneously. Each has an owner, contract and observable acceptance evidence. One implementation exists per capability.

Order	Client outcome and canonical seam	Acceptance evidence and boundary
0: completed locally	Loaded storage objects and generated IVDR artifacts match their recorded identity. `storage-provider.ts`, local/S3 providers and `ivdr-binder-routes.ts`.	106 focused passing tests with fail-first corruption cases; scoped typecheck/lint; independent review. No deployed durability or agency acceptance claim.
1: classification completed locally; qualification remains	External-validator classification and actual advisory outcomes now agree in the existing gate, assessor, screens, QC, pre-transmit and MCP. Official-format currency, qualified package receipts and delivery remain separate work.	230 passing focused tests and 34 independent checks. Clean subset cannot clear required external validation. Matching qualified reports still need package/report hashes, engine/profile/version, authority and coverage before qualification is claimed.
2: recovery completed locally; operational qualification remains	Uploads remain visible and repairable after failed extraction/catalog persistence. Existing capture and `data-room-processing.ts` paths.	Fail-first recovery cases now pass, with exact original binding, fenced retries, honest persisted-text views and real PostgreSQL concurrency/withdrawal checks. No client backlog or scheduler was activated; deployed workload qualification remains.
3: identity foundation advanced; original anchors remain	Original bytes, extracted text and page/table anchors remain distinguishable. Existing capture, processor and chunking records.	Same-length re-extraction invalidates old anchors; table/cell/page locators resolve; partial/OCR/translation uncertainty is disclosed. Family identity comes from stable origin/check-in or reviewed matching, rather than title alone.
4: preview completed locally; fact decisions remain	The read-only preview checks framed source statements; a future reviewer can inspect, accept, reject or correct independently persisted facts. Existing catalog core, governance service, tools and Vault panel.	Wrong-assay number, measured-result/specification swap, wrong specimen/method/unit/denominator and incomplete context refuse or stay unresolved. Per-fact revisions and decisions survive retries and source changes; real PostgreSQL access/race tests and reviewer workflow pass.
5: relevant evidence for AnA	Drafting retrieves complete-context passages anywhere in selected eligible documents. Existing Vault index/RAG, loader and authoring verification.	Late table beyond character 80,000 is retrieved with headers/footnotes. Versioned multi-range receipts round-trip through generation/save/export; stale source, changed extraction and revoked project access refuse. Ten specified failure fixtures and legacy-prefix compatibility pass.
6: reproducible calculated claims	One admitted calculation links its inputs, approved analysis policy and exact output to a document span. Existing computation runs, method/replay registry and span lineage.	Independent reference values and inclusion/exclusion rules match; original inputs and method versions reproduce the result; unsupported designs refuse. Begin with a qualified pharma or IVD method on existing admitted formats.
7: reviewed change consequences	Proposed source, cutoff, configuration or rule changes identify affected scientific claims and review tasks. Existing source-usage, derived-impact, comparator and governed actions.	Critical affected nodes found; unaffected controls preserved; stale preview cannot apply; no silent edits to approved content. Historical package remains reconstructible.

Order	Client outcome and canonical seam	Acceptance evidence and boundary
8: scoped filing qualification	One application/pathway at a time gains a verified outline, package, validator and applicant channel. Existing regional packager, official-template engines and gateway owners.	Official golden artifact, local expert scope review, independent validator receipt and honest handoff/delivery state. Accepted transmission requires real authorized channel evidence. See F0–F9 for ownership and fixtures.

Priority 1 began as a **confirmed component finding**, not a full submission bypass. The original probe showed clean subset clearance without a licensed validator; the current portable regression shows required clearance false. The continued tests also exercise the actual assessor with controlled DB boundaries and preserve composed release gates. Transport and real database/provider qualification remain untested. Baseline/current probe and exact scope. A configurable LORENZ adapter and report import already exist; further qualification should extend these rather than create a parallel validator subsystem.

Scientific facts and AI behavior

Start with five discriminated families: pharma clinical results; CMC results/specifications; device test results/acceptance criteria; IVD diagnostic performance; and IVD analytical performance. A source-stated fact and a server-computed result have different origins. Every fact carries its complete candidate revision, exact witnesses, original/extraction identity, evaluation and attributable human decision.

Review a claim's transcription, contextual agreement and scientific applicability separately. Missing denominator, reference standard, specimen or assay identity remains unresolved. Comparator agreement is not automatically sensitivity/specificity. A measured test result cannot inherit a specification's label. OCR confidence is not original-page confirmation. Preserve real legacy catalog confirmations while explicitly stating that their property/context binding was not recorded; never silently reinterpret an old approval as typed scientific verification.

Useful new AI capabilities include constrained structured proposals, relevant multi-range retrieval, original-page/table inspection, bilingual source/translation comparison, bounded tool use, and explanations of deterministic differences. Qualify each for its declared purpose. Multimodal inspection can propose evidence regions; it does not establish a calculation, scientific conclusion or regulatory decision by itself.

Reuse the existing model gateway and approved model registry. Version and evaluate model, prompt, retrieval, extraction and rule changes together for the affected cohort. Client placement, retention, purpose and project rights apply before model context is sent. A provider outage cannot relax those terms. Long-running work should checkpoint successful sections, propagate cancellation and fence retries within existing AnA run/job mechanisms; no new orchestration platform is justified by this study.

Market and lifecycle delivery

All requested sectors and markets are research targets. Installed metadata, a retained engine or a generated ZIP alone does not establish a qualified customer workflow. The detailed matrix separately records authoring, artifact production, independent validation, template and transport.

Market	Medicine/biotechnology priorities	Device/diagnostic priorities
USA	Qualify one real IND sequence and current validation/channel contract; separate new-application eCTD4 from v3 lifecycle. Biologic/CGT CMC, BLA and safety obligations need their own scoped packs.	Qualify current official nIVD/IVD eSTAR and attachment behavior on one 510(k) pathway; then exact De Novo/PMA/request scopes. Vendored 7.0 versus recorded current 7.1 requires truthful format policy.

Market	Medicine/biotechnology priorities	Device/diagnostic priorities
EU/EEA	CTIS trial evidence and country-specific Part II; distinguish centralised marketing from national/MRP/DCP. IRIS manages the new iMAA procedure, while the dossier goes through eSubmission Gateway.	Retain existing Annex II/III technical-file export, then qualify reachable MDR/IVDR workflows and NB handoff. IVD scientific validity, analytical and clinical performance are separate evidence tracks. EUDAMED registration is a separate obligation/channel.
UK	Combined IRAS trial preparation under effective rules, with dated transition decisions; marketing recognition and lifecycle need separate MHRA packs.	GB and NI classification, conformity, registration, study and PMS/MORE branches remain separate. EU technical files do not automatically complete UK obligations.
Japan	Japanese regional v4 medicine-approval scope, vocabulary and bilingual signoff; trial notification and regenerative-product requirements separately classified.	Product/class-specific approval, certification or notification with local MAH/DMAH and performance evidence. Medicine eCTD4 rules do not automatically govern devices/IVDs.
Canada	Activity-specific eCTD/non-eCTD/REP and Canadian M1; CTA, Schedule D biologic, site/lot and lifecycle obligations separately scoped.	Class-specific MDL/ITA and local evidence/ToC requirements; current availability of any limited eSTAR programme must be verified before offering it.
Australia	Current AU M1 and medicine lifecycle; distinguish prescription biological medicines from the legal biologicals framework before selecting trial/dossier/biovigilance obligations.	ARTG/sponsor/classification, Essential Principles, conformity, clinical/performance evidence, changes and vigilance. No drug eCTD default for device registration.

Each route pack should encode product/class, pathway and lifecycle stage; binding law versus guidance; official source and retained version; effective and transition dates; local language/entity; applicable evidence; format/template; channel; and platform qualification. Preserve historical rules and decisions instead of rewriting previous filings when guidance changes. Q0–Q4 progression in the regulatory specification separates scoped drafts, expert-reviewed requirements, qualified deterministic exports, authorised official-channel tests and sponsor-controlled operational use.

Current launch boundaries remain explicit. Native XPT/SAS/Dataset-JSON readers, dataset-bundle ZIP intake and m5 dataset delivery are deferred. CMC composition currently supports small-molecule Module 3 for a US IND and refuses wider scopes. Hidden device-kit activation and new regional composition/transmission belong to their recorded product/release workstreams. This plan prepares those increments without removing refusals or claiming they shipped.

Proving a moat

Evaluate five concrete hypotheses: scientifically justified reuse and bridging; reproducible exclusions/denominators/units; precise pre-change impact; historical filing reconstruction; and reviewed corrections that become regression evidence. These may outperform a specific workflow even where a competitor advertises a similar category. Public product pages cannot establish competitive absence.

The proposed full benchmark contains **90 cases**, from three product archetypes × five requested market groups × six variants, with a separate UK extension. This is a protocol, not an executed benchmark or reliability estimate. Begin with three expert-labelled science workflows and the customer's current process; then compare relevant licensed workflows from Weave, Veeva/ArisGlobal/Certara, Rimsys, Greenlight Guru/MasterControl or Benchling as applicable.

Freeze independent expected source anchors, frames, calculations, reuse decisions, affected/unaffected nodes and route boundaries before runs. Use clean, lookalike, late-table, change, historical-rule and

revoked/stale-access variants. Measure critical unsupported claims, useful coverage, calculation disagreement, impact precision/recall, reconstruction, active SME review time, retries and total workflow cost. Report unavailable competitor capabilities as not assessed. Japanese outputs need scientific and bilingual review.

Build the context/decision/lineage semantics and evaluation corpus. Reuse existing canonical catalog, computation, authoring, lifecycle and gateway services. Buy or integrate qualified publishing/validation, OCR/table extraction and existing customer systems when they solve a measured problem. Start with reliable source connectors and reviewed publishing handoff; retain customer-owned relationships and portable exports. Cross-client improvement uses public, licensed or explicitly permitted evidence rather than pooling private dossiers.

Implementation and release discipline

Publish the shared interface and meaningful failing examples before parallel implementation. Divide ownership by subsystem on `concept2cure-v2`, then use an independent reviewer and held-out cases. Prefer state-sequence, differential, transaction and fault-injection tests over tests that merely repeat implementation details. A gate must be observed refusing the bad case before being credited as effective.

New tenant content needs replay-safe migrations, public/integer organization identity, RLS/access proof, append-only privileges where promised, and export/purge integration. Experimental paths remain off until their corpus, database, UI and operational checks pass. Extend existing launch rows with exact evidence; do not declare D1–D10 green from this local patch. Full application builds, live storage recovery, real PostgreSQL isolation, installed rule packs, provider qualification and genuine agency acknowledgements require their own evidence.

The next source-read increment should define and qualify exact dispatch and replay checkpoints, then carry admitted nonzero/multiple ranges through generation, save and export. Actor-filtered discovery/recall and exact-source host-admission metadata provide the local foundation; they do not establish provider delivery or task-specific comprehension. The next scientific increment should define immutable per-fact decision storage and lifecycle/export/purge behavior, then qualify a client-expert-labelled cohort before enabling acceptance. The completed preview is a bounded source check, not that approval workflow. Qualified external package receipts remain a separate D7 increment. Competitive pilots should assess the resulting client workflow before broader adapters or infrastructure are enabled.